Introduction

We might have come across the word "**Database**" quite often. This term carries a high emphasis on its arms. More often, it is not just related to the developer's perspective but is quite often used with non-tech groups or communities. Technically, a database is more of a storage term used to denote the relationship with different forms of data that are coagulated in a single place. Thus, we can define a database as an organized collection of data, generally stored and accessed electronically through computer systems.

A good database design starts with a list of the data that you want to include in your database and what you want to be able to do with the database later on. This can all be written in your own language, without any SQL. In this stage you must try not to think in tables or columns, but just think: "What do I need to know?" Don't take this too lightly, because if you find out later that you forgot something, usually you need to start all over. Adding things to your database is mostly a lot of work.

What is Database Design?

Database design can be generally defined as a collection of tasks or processes that enhance the designing, development, implementation, and maintenance of enterprise data management system. Designing a proper database reduces the maintenance cost thereby improving data consistency and the cost-effective measures are greatly influenced in terms of disk storage space. Therefore, there has to be a brilliant concept of designing a database. The designer should follow the constraints and decide how the elements correlate and what kind of data must be stored.

The main objectives behind database designing are to produce physical and logical design models of the proposed database system. To elaborate this, the logical model is primarily concentrated on the requirements of data and the considerations must be made in terms of monolithic considerations and hence the stored physical data must be stored independent of the physical conditions. On the other hand, the physical database design model includes a translation of the logical design model of the database by keep control of physical media using hardware resources and software systems such as Database Management System (DBMS).

Why is Database Design important?

The important consideration that can be taken into account while emphasizing the importance of database design can be explained in terms of the following points given below.

1. Database designs provide the blueprints of how the data is going to be stored in a system. A proper design of a database highly affects the overall performance of any application.
2. The designing principles defined for a database give a clear idea of the behavior of any application and how the requests are processed.
3. Another instance to emphasize the database design is that a proper database design meets all the requirements of users.

4. Lastly, the processing time of an application is greatly reduced if the constraints of designing a highly efficient database are properly implemented.

Life Cycle

Before jumping directly on the designing models constituting database design it is important to understand the overall workflow and life-cycle of the database.

Requirement Analysis

First of all, the planning has to be done on what are the basic requirements of the project under which the design of the database has to be taken forward. Thus, they can be defined as:-

Planning - This stage is concerned with planning the entire DDLC (Database Development Life Cycle). The strategic considerations are taken into account before proceeding.

System definition - This stage covers the boundaries and scopes of the proper database after planning.

Database Designing

The next step involves designing the database considering the user-based requirements and splitting them out into various models so that load or heavy dependencies on a single aspect are not imposed. Therefore, there has been some model-centric approach and that's where logical and physical models play a crucial role.

Physical Model - The physical model is concerned with the practices and implementations of the logical model.

Logical Model - This stage is primarily concerned with developing a model based on the proposed requirements. The entire model is designed on paper without any implementation or adopting DBMS considerations.

Implementation

The last step covers the implementation methods and checking out the behavior that matches our requirements. It is ensured with continuous integration testing of the database with different data sets and conversion of data into machine understandable language. The manipulation of data is primarily focused on these steps where queries are made to run and check if the application is designed satisfactorily or not.

Data conversion and loading - This section is used to import and convert data from the old to the new system.

Testing - This stage is concerned with error identification in the newly implemented system. Testing is a crucial step because it checks the database directly and compares the requirement specifications.

Database Design Process

The process of designing a database carries various conceptual approaches that are needed to be kept in mind. An ideal and well-structured database design must be able to:

1. Save disk space by eliminating redundant data.
2. Maintains data integrity and accuracy.
3. Provides data access in useful ways.
4. Comparing Logical and Physical data models.

Logical

A logical data model generally describes the data in as many details as possible, without having to be concerned about the physical implementations in the database. Features of logical data model might include:

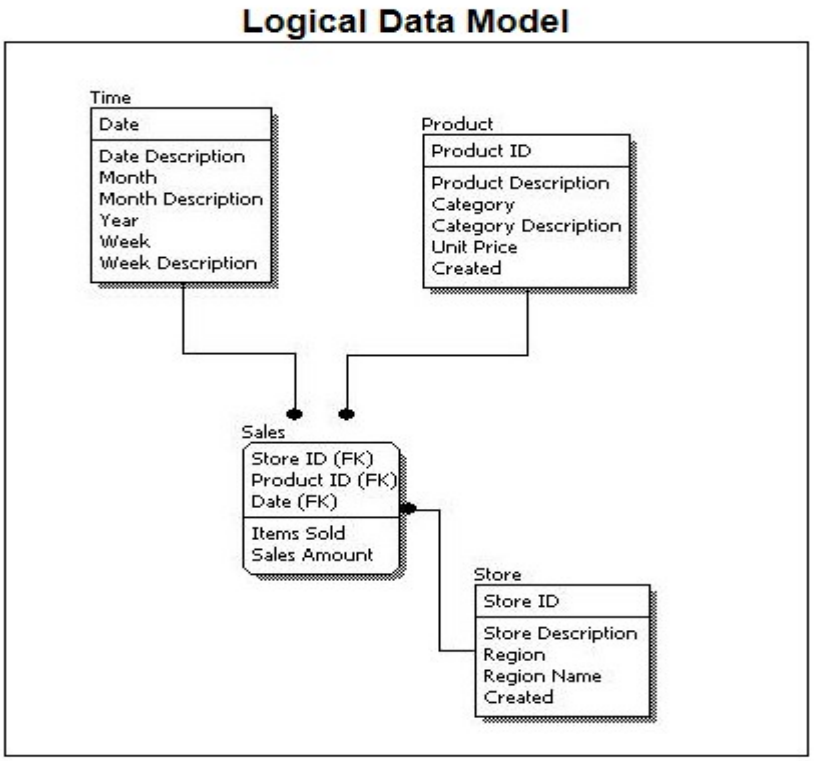
1. All the entities and relationships amongst them.
2. Each entity has well-specified attributes.
3. The primary key for each entity is specified.
4. Foreign keys which are used to identify a relationship between different entities are specified.
5. Normalization occurs at this level.

A logical model can be designed using the following approach:

1. Specify all the entities with primary keys.
2. Specify concurrent relationships between different entities.
3. Figure out each entity attributes
4. Resolve many-to-many relationships.
5. Carry out the process of normalization.

Also, one important factor after following the above approach is to critically examine the design based on requirement gathering. If the above steps are strictly followed, there are chances of creating a highly efficient database design that follows the native approach.

To understand these points, see the image below to get a clear picture.



If we compare the logical data model as shown in the figure above with some sample data in the diagram, we can come up with facts that in a conceptual data model there are no presence of a primary key whereas a logical data model has primary keys for all of its attributes. Also, logical data model the cover relationship between different entities and carries room for foreign keys to establish relationships among them.

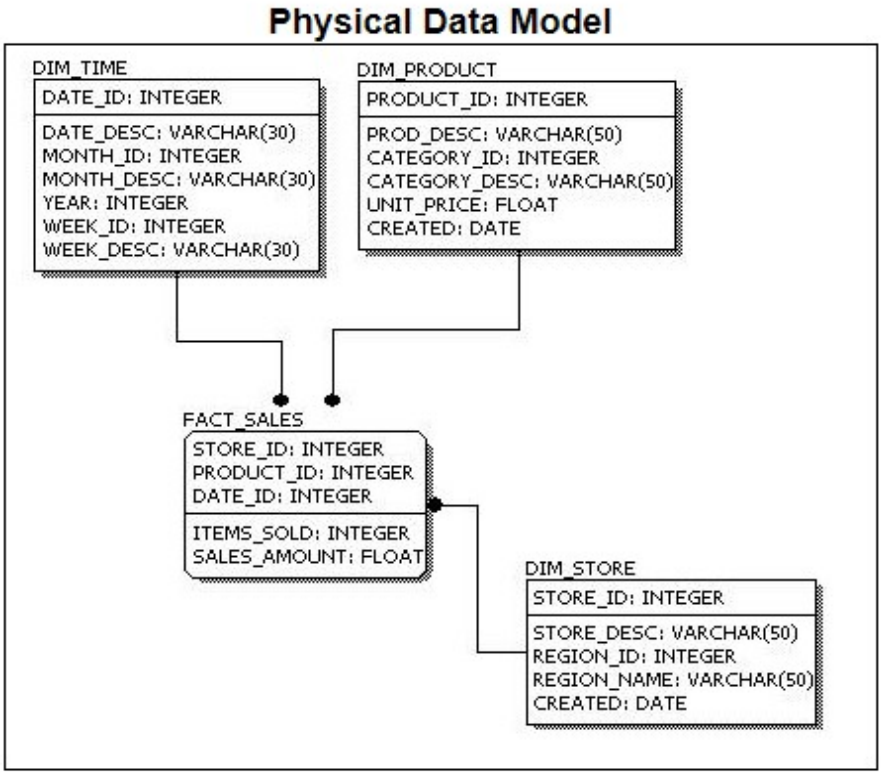
Physical

A Physical data mode generally represents how the approach or concept of designing the database. The main purpose of the physical data model is to show all the **structures** of the table including the **column name, column data type, constraints, keys(primary and foreign)**, and the relationship among tables. The following are the features of a physical data model:

1. Specifies all the columns and tables.
2. Specifies foreign keys that usually define the relationship between tables.
3. Based on user requirements, de-normalization might occur.
4. Since the physical consideration is taken into account so there will straightforward reasons for difference than a logical model.
5. Physical models might be different for different RDBMS. For example, the data type column may be different in MySQL and SQL Server.

While designing a physical data model, the following points should be taken into consideration:

1. Convert the entities into tables.
2. Convert the defined relationships into foreign keys.
3. Convert the data attributes into columns.
4. Modify the data model constraints based on physical requirements.



Comparing this physical data model with the logical with the previous logical model, we might conclude the differences that in a physical database entity names are considered table names and attributes are considered column names. Also, the data type of each column is defined in the physical model depending on the actual database used.

Identifying Entities

The types of information that are saved in the database are called 'entities'. These entities exist in four kinds: people, things, events, and locations. Everything you could want to put in a database fits into one of these categories. If the information you want to include doesn't fit into these categories then it is probably not an entity but a property of an entity, an attribute. For example. Imagine that you are creating a website for a shop, what kind of information do you have to deal with? In a shop you sell your products to customers. The "Shop" is a location; "Sale" is an event; "Products" are things; and "Customers" are people. These are all entities that need to be included in your database.

But what other things are happening when selling a product? A customer comes into the shop, approaches the vendor, asks a question and gets an answer. "Vendors" also participate, and because vendors are people, we need a vendors entity.

Figure 3: Entities: types of information.

Identifying Relationships

The next step is to determine the relationships between the entities and to determine the cardinality of each relationship. The relationship is the connection between the entities, just like in the real world: what does one entity do with the other, how do they relate to each other? For example, customers buy products, products are sold to customers, a sale comprises products, a sale happens in a shop.

The cardinality shows how much of one side of the relationship belongs to how much of the other side of the relationship. First, you need to state for each relationship, how much of one side belongs to exactly 1 of the other side. For example: How many customers belong to 1 sale?; How many sales belong to 1 customer?; How many sales take place in 1 shop?

You'll get a list like this: (please note that 'product' represents a type of product, not an occurance of a product)

- Customers --> Sales; 1 customer can buy something several times
- Sales --> Customers; 1 sale is always made by 1 customer at the time
- Customers --> Products; 1 customer can buy multiple products
- Products --> Customers; 1 product can be purchased by multiple customers
- Customers --> Shops; 1 customer can purchase in multiple shops
- Shops --> Customers, 1 shop can receive multiple customers
- Shops --> Products; in 1 shop there are multiple products
- Products --> Shops; 1 product (type) can be sold in multiple shops
- Shops --> Sales; in 1 shop multiple sales can me made
- Sales --> Shops; 1 sale can only be made in 1 shop at the time
- Products --> Sales; 1 product (type) can be purchased in multiple sales
- Sales --> Products; 1 sale can exist out of multiple products

Did we mention all relationships? There are four entities and each entity has a relationship with every other entity, so each entity must have three relationships, and also appear on the left end of the relationship three times. Above, 12 relationships were mentioned, which is 4*3, so we can conclude that all relationships were mentioned.

Now we'll put the data together to find the cardinality of the whole relationship. In order to do this, we'll draft the cardinalities per relationship. To make this easy to do, we'll adjust the notation a bit, by noting the 'backward'-relationship the other way around:

- Customers --> Sales; 1 customer can buy something several times
- Sales --> Customers; 1 sale is always made by 1 customer at the time

The second relationship we will turn around so it has the same entity order as the first. Please notice the arrow that is now faced the other way!

- Customers <-- Sales; 1 sale is always made by 1 customer at the time

Cardinality exists in four types: one-to-one, one-to-many, many-to-one, and many-to-many. In a database design this is indicated as: 1:1, 1:N, M:1, and M:N. To find the right indication just leave the '1'. If there is a 'many' on the left side, this will be indicated with 'M', if there is a 'many' on the right side it is indicated with 'N'.

- Customers --> Sales; 1 customer can buy something several times; 1:N.
- Customers <-- Sales; 1 sale is always made by 1 customer at the time; 1:1.

The true cardinality can be calculated through assigning the biggest values for left and right, for which 'N' or 'M' are greater than '1'. In thisexample, in both cases there is a '1' on the left side. On the right side, there is a 'N' and a '1', the 'N' is the biggest value. The total cardinality is therefore '1:N'. A customer can make multiple 'sales', but each 'sale' has just one customer.

If we do this for the other relationships too, we'll get:

- Customers --> Sales; --> 1:N
- Customers --> Products; --> M:N
- Customers --> Shops; --> M:N
- Sales --> Products; --> M:N
- Shops --> Sales; --> 1:N
- Shops --> Products; --> M:N

So, we have two '1-to-many' relationships, and four 'many-to-many' relationships.

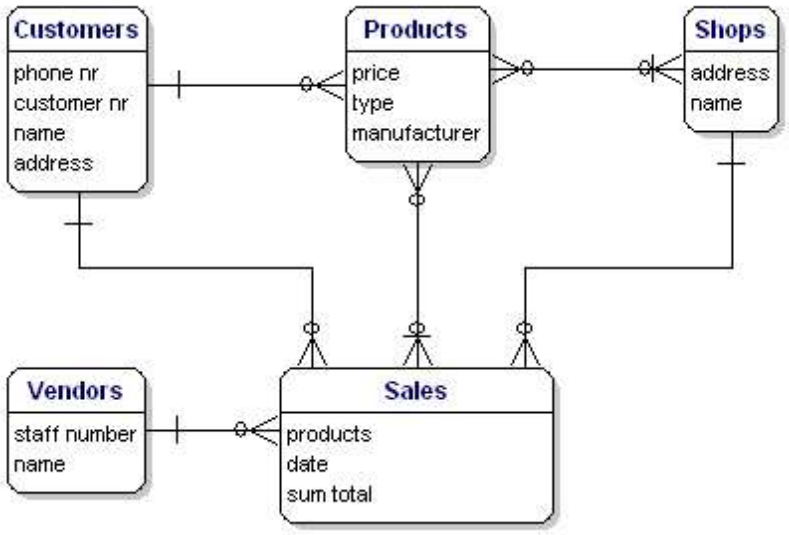


Figure 4: Relationships between the entities.

Between the entities there may be a mutual dependency. This means that the one item cannot exist if the other item does not exist. For example, there cannot be a sale if there are no customers, and there cannot be a sale if there are no products.

The relationships Sales --> Customers, and Sales --> Products are mandatory, but the other way around this is not the case. A customer can exist without sale, and also a product can exist without sale. This is of importance for the next step.

Recursive Relationships

Sometimes an entity refers back to itself. For example, think of a work hierarchy: an employee has a boss; and the bosschef is an employee too. The attribute 'boss' of the entity 'employees' refers back to the entity 'employees'.

In an ERD this type of relationship is a line that goes out of the entity and returns with a nice loop to the same entity.

Redundant Relationships

Sometimes in your model you will get a 'redundant relationship'. These are relationships that are already indicated by other relationships, although not directly.

In the case of our example there is a direct relationships between customers and products. But there are also relationships from customers to sales and from sales to products, so indirectly there already is a relationship between customers and products through sales. The relationship 'Customers <----> Products' is made twice, and one of them is therefore redundant. In this case,

products are only purchased through a sale, so the relationships 'Customers <----> Products' can be deleted. The model will then look like this:

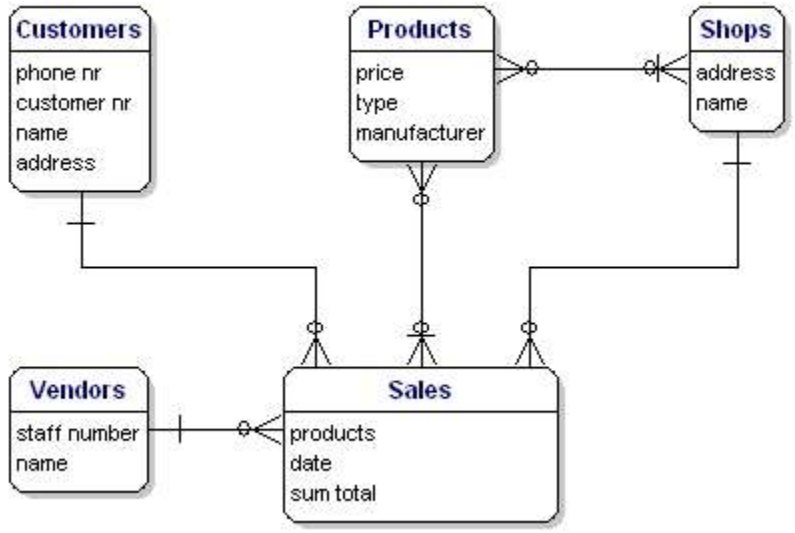


Figure 5: Relationships between the entities.

Solving Many-to-Many Relationships

Many-to-many relationships (M:N) are not directly possible in a database. What a M:N relationship says is that a number of records from one table belongs to a number of records from another table. Somewhere you need to save which records these are and the solution is to split the relationship up in two one-to-many relationships.

This can be done by creating a new entity that is in between the related entities. In our example, there is a many-to-many relationship between sales and products. This can be solved by creating a new entity: sales-products. This entity has a many-to-one relationship with Sales, and a many-to-one relationship with Products. In logical models this is called an associative entity and in physical database terms this is called a link table, intersection table or junction table.

Figure 6: Many to many relationship implementation via associative entity.

In the example there are two many-to-many relationships that need to be solved: 'Products <----> Sales', and 'Products <----> Shops'. For both situations there needs to be created a new entity, but what is that entity?

For the Products <----> Sales relationship, every sale includes more products. The relationship shows the content of the sale. In other words, it gives details about the sale. So the entity is called 'Sales details'. You could also name it 'sold products'.

The Products <----> Shops relationship shows which products are available in which the shops, also known as 'stock'. Our model would now look like this:

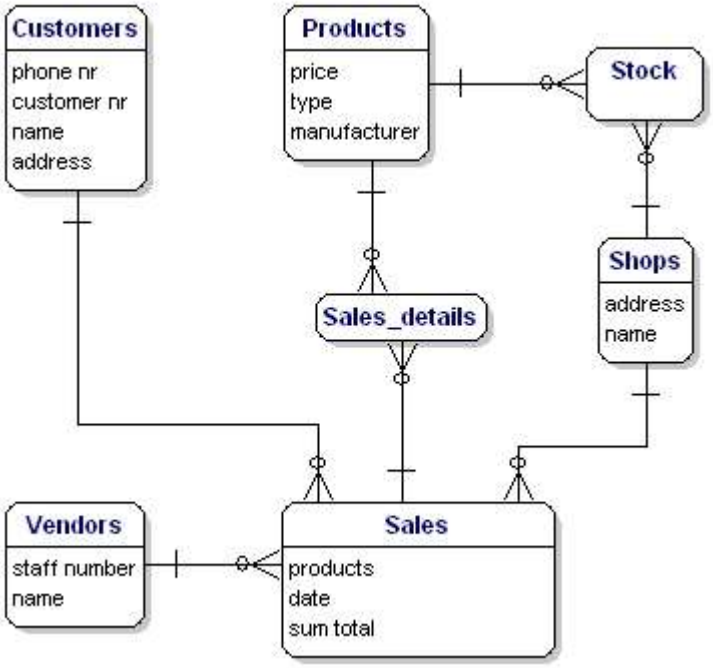


Figure 7: Model with link tables Stock and Sales_details.

Identifying Attributes

The data elements that you want to save for each entity are called 'attributes'.

About the products that you sell, you want to know, for example, what the price is, what the name of the manufacturer is, and what the type number is. About the customers you know their customer number, their name, and address. About the shops you know the location code, the name, the address. Of the sales you know when they happened, in which shop, what products were sold, and the sum total of the sale. Of the vendor you know his staff number, name, and address. What will be included precisely is not of importance yet; it is still only about what you want to save.

Figure 8: Entities with attributes.

Derived Data

Derived data is data that is derived from the other data that you have already saved. In this case the 'sum total' is a classical case of derived data. You know exactly what has been sold and what each product costs, so you can always calculate how much the sum total of the sales is. So really it is not necessary to save the sum total.

So why is it saved here? Well, because it is a sale, and the price of the product can vary over time. A product can be priced at 10 euros today and at 8 euros next month, and for your administration you need to know what it cost at the time of the sale, and the easiest way to do this is to save it here

Presenting Entities and Relationships: Entity Relationship Diagram (ERD)

The Entity Relationship Diagram (ERD) gives a graphical overview of the database. There are several styles and types of ER Diagrams. A much-used notation is the 'crowfeet' notation, where entities are represented as rectangles and the relationships between the entities are represented as lines between the entities. The signs at the end of the lines indicate the type of relationship. The side of the relationship that is mandatory for the other to exist will be indicated through a dash on the line. Not mandatory entities are indicated through a circle. "Many" is indicated through a 'crowfeet'; the relationship-line splits up in three lines.

A 1:1 mandatory relationship is represented as follows:

Figure 9: Mandatory one to one relationship.